nature genetics

Control of root system architecture by *DEEPER ROOTING 1* increases rice yield under drought conditions

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The genetic improvement of drought resistance is essential for stable and adequate crop production in drought-prone areas[1]. Here we demonstrate that alteration of root system architecture improves drought avoidance through the cloning and characterization of *DEEPER ROOTING 1* (*DRO1*), a rice quantitative trait locus controlling root growth angle. *DRO1* is negatively regulated by auxin and is involved in cell elongation in the root tip that causes asymmetric root growth and downward bending of the root in response to gravity. Higher expression of *DRO1* increases the root growth angle, whereby roots grow in a more downward direction. Introducing *DRO1* into a shallow-rooting rice cultivar by backcrossing enabled the resulting line to avoid drought by increasing deep rooting, which maintained high yield performance under drought conditions relative to the recipient cultivar. Our experiments suggest that control of root system architecture will contribute to drought avoidance in crops.

The world population is expected to reach 9 billion by the middle of the twenty-first century. Given the projected population increase, a 40% improvement in crop yields in drought-prone areas is needed by 2025 (ref. 1). Without irrigation, however, increasing crop production under drought conditions is difficult. Rice, a staple food for nearly half of the world's population, is particularly susceptible to drought-induced stress owing to its shallow rooting relative to other cereal crops[2]. Deep rooting may help plants to avoid drought-induced stress by extracting water from deep soil layers[3–5]. To improve drought avoidance in rice, therefore, introducing the deep-rooting characteristic into shallow-rooting cultivars is considered one of the most promising breeding strategies[5]. So far, no quantitative trait loci (QTLs) responsible for deep rooting have been cloned in rice[6,7], although some root QTLs have recently been mapped[8–12]. To improve drought avoidance in rice, identifying genes influencing deep rooting and clarifying their effects on drought avoidance is necessary.

Cultivated rice lines vary greatly in root system architecture[13]. We selected two distinct accessions in a preliminary screen for natural variation in root system architecture[10]. IR64, a leading paddy cultivar widely grown in Asia, showed shallow rooting in an upland field, whereas Kinandang Patong (KP), an upland cultivar from the Philippines, showed deep rooting (**Supplementary Fig. 1**). We previously mapped a QTL for deep rooting, *DRO1*, to chromosome 9 (**Fig. 1a**, top) in the progeny of a IR64 × KP cross[10]. Here we developed a near-isogenic line homozygous for the KP allele of *DRO1* (*DRO1-kp*) in the IR64 genetic background (Dro1-NIL) to compare the effects of *DRO1-kp* and the IR64 allele of *DRO1* (*DRO1-ir*) on deep rooting under field conditions. The maximum root depth of Dro1-NIL plants was more than twice that of IR64 plants (**Fig. 1a**, bottom and **Supplementary Fig. 1**). Dro1-NIL plants had more roots distributed in deeper soil layers than did IR64 plants (**Supplementary Figs. 2** and **3a**). Notably, Dro1-NIL plants did not show any significant differences from IR64 plants in total-root dry weight or in shoot traits, indicating that the *DRO1-kp* allele changed only the root distribution in the soil (**Supplementary Fig. 3b**). Deep rooting in cereals is a complex trait consisting of the root growth angle and maximum root length[14]. The root growth angle (a measure of downward growth) of Dro1-NIL plants was approximately twice that of IR64 plants (**Supplementary Fig. 4**). Dro1-NIL plants showed slight differences in root length from IR64 plants but no marked differences in other root and shoot traits (**Supplementary Fig. 5**). These results suggest that *DRO1* mainly influences root growth angle.

We performed positional cloning of *DRO1*. High-resolution mapping narrowed the candidate region to a 6.0-kb segment (**Supplementary Fig. 6**). One putative ORF, for Os09g0439800 (LOC_Os09g26840.1), is annotated in this region (Rice Annotation Project Database (RAP-DB; see URLs) and **Fig. 1b**). Comparing the sequence of this ORF in the IR64 and KP accessions identified a single 1-bp deletion within exon 4 in IR64 (**Fig. 1b**), resulting in the introduction of a premature stop codon. Transforming an 8.7-kb genomic fragment

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Received 13 March; accepted 15 July; published online 4 August 2013; doi:10.1038/ng.2725

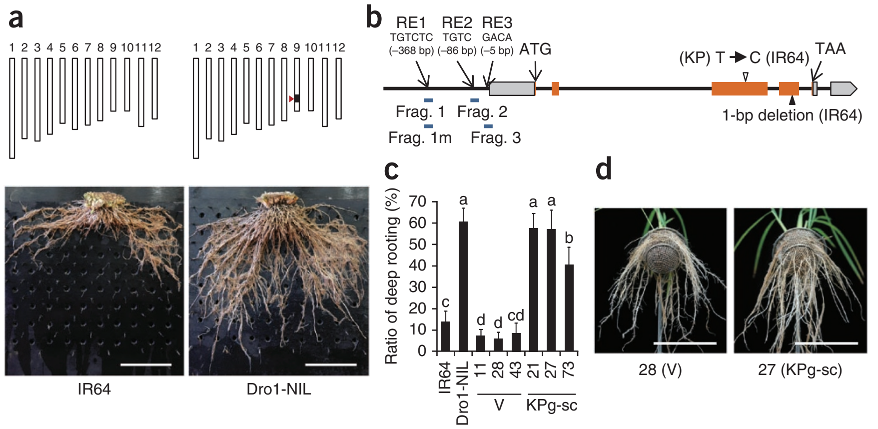


Figure 1 Phenotypic characterization and cloning of *DRO1*. (**a**) Graphical genotypes (chromosomes 1–12) and vertical root distribution of IR64 and Dro1-NIL. White rectangles, homozygous regions from IR64; black box, homozygous region from KP; red arrowhead, position of *DRO1*. Roots of each line were obtained from an upland field using a monolith sampler (30 cm × 30 cm × 5 cm). Scale bars, 10 cm. (**b**) Structure of *DRO1* based on annotation of Os09g0439800 (LOC_Os09g26840.1). Orange rectangles, ORF; gray rectangles, 5′ and 3′ UTRs; white arrowhead, synonymous nucleotide substitution; black arrowhead, single-nucleotide deletion in IR64 relative to KP; RE1, RE2 and RE3, AuxREs in the *DRO1* promoter region (numbers in parentheses show the position of each element relative to the start of the 5′ UTR); blue bars, fragments (frag.) used as probes for the electrophoretic mobility shift assay (**Fig. 4b**). (**c**) Ratio of deep rooting of transgenic IR64 (T_2 plants) containing a single copy of the 8.7-kb *DRO1* genomic fragment from KP (KPg-sc) or the empty vector (V). Data are shown as mean ± s.d.; $n = 20$ plants for all lines except for V 11, for which $n = 16$. Values labeled with different letters differ significantly among the eight lines ($P < 0.05$, Tukey's honestly significant difference test). (**d**) Basket assay of representative transgenic plants containing vector control and a single copy of the 8.7-kb *DRO1* genomic fragment from KP in the IR64 genetic background (20 plants in each line). Scale bars, 10 cm.

of KP containing the candidate gene into IR64 increased the ratio of deep rooting (an index of downward root growth; Online Methods) compared to IR64 transformed with the vector control (**Fig. 1c**,**d**).

The predicted protein product of *DRO1* has no similarity to known proteins. Genes found in monocots had higher homology to *DRO1* than genes found in dicots (**Supplementary Fig. 7**). To determine the subcellular localization of the DRO1 protein, we introduced a construct encoding a DRO1-kp–YFP (yellow fluorescent protein) or DRO1-ir–YFP fusion protein into rice protoplasts. Fluorescence of DRO1-kp–YFP was localized to the plasma membrane (**Supplementary Fig. 8**). *DRO1* is not predicted to encode a signal peptide or a transmembrane domain but is predicted to encode two putative N-myristoylation sites associated with lipid modification (**Supplementary Fig. 7**), suggesting that DRO1 is associated with a membrane protein or with the plasma membrane. In contrast, the *DRO1-ir*::*YFP* construct produced a truncated protein that was localized to the nucleus and cytoplasm as well as to the plasma membrane (**Supplementary Fig. 8**). The C-terminal sequence of DRO1, which is highly conserved in terrestrial plants (**Supplementary Fig. 7**), might be associated with the stability of the plasma membrane localization of DRO1. Because our results were obtained from transient expression assays, further analysis will be needed to clarify the subcellular localization of DRO1 in rice.

To understand the mechanisms underlying the *DRO1*-controlled phenotype, we investigated root growth angle in the seminal roots of IR64, Dro1-NIL and several *DRO1* transgenic lines. IR64, vector control lines and RNA interference (RNAi) knockdown lines had a smaller root growth angle and higher variation in this trait than Dro1-NIL and the *DRO1-kp* transgenic lines, suggesting that *DRO1-ir* and RNAi knockdown lines have less gravitropic curvature than *DRO1-kp* lines (**Fig. 2a**,**b** and **Supplementary Figs. 9a**,**b** and **10a**,**b**). Gravitropism is one of the most important factors determining root growth angle[15]. A time course of root gravitropic curvature showed that the roots of Dro1-NIL plants responded more abruptly to rotation from the normal vertical axis to the horizontal axis than did those of IR64 plants (**Fig. 2c**,**d**). We also examined root gravitropic curvature at different angles from the normal vertical axis. Roots of Dro1-NIL and the *DRO1-kp* transgenic lines responded more sharply at larger inclination

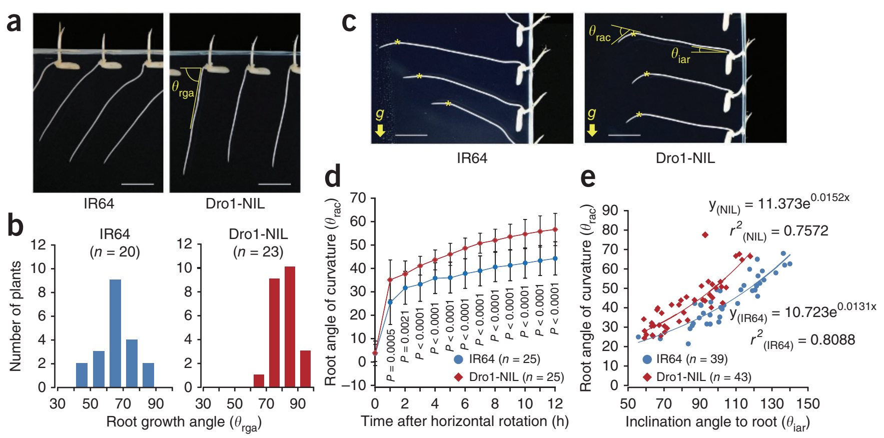
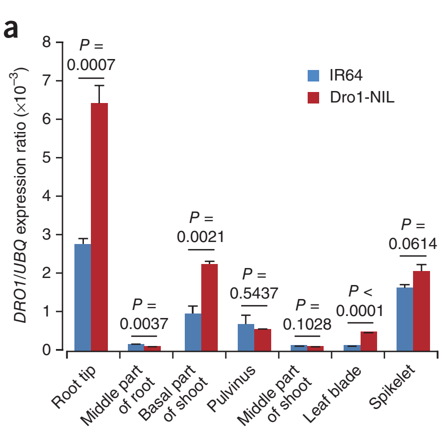


Figure 2 Effect of *DRO1* on root growth angle and root gravitropic curvature. (**a**) Root growth angle (θ_{rga}) of IR64 and Dro1-NIL plants 2 d after sowing in agarose gel. Scale bars, 1 cm. (**b**) Distribution of root growth angle in IR64 and Dro1-NIL plants. IR64, mean ± s.d. = 65.2° ± 11.4°; Dro1-NIL, mean ± s.d. = 80.9° ± 7.1°. (**c**) Gravitropic curvature in seminal roots of IR64 and Dro1-NIL plants. Seedlings were grown on agarose for 2 d and then rotated either 60° or 90° from the original vertical axis for 4 h (Online Methods). θ_{iar}, inclination angle of root, representing the angle between the root and the horizontal axis immediately after rotation; θ_{rac}, root angle of curvature after rotation. Asterisks indicate the positions of root tips at the start of rotation. Yellow arrows indicate the direction of gravitational force. Scale bars, 1 cm. (**d**) Time course of the root angle of curvature after horizontal rotation in IR64 and Dro1-NIL plants. Seedlings were kept in the dark after rotation, and curvature was measured once per hour for 12 h. Data are shown as mean ± s.d. *P* values are based on Student's *t* tests. (**e**) Relationship between the inclination angle of the root and the root angle of curvature after rotation in IR64 and Dro1-NIL plants. *y*, root angle of curvature; *x*, inclination angle to root; r^2, ratio of contribution.



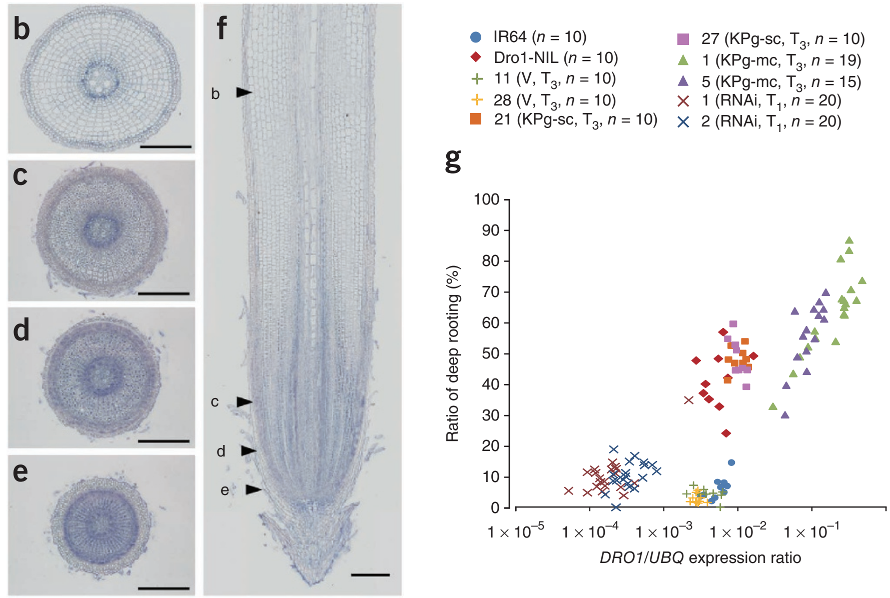


Figure 3 Expression analysis of *DRO1*. (**a**) *DRO1* expression in several shoot and root tissues. Expression data for *DRO1* were normalized to the expression of a ubiquitin gene. Samples of the root tip, the middle part of the root, the basal part of the shoot (1-cm sample from the bottom of the shoot, including the meristems of adventitious root primordia), the middle part of the shoot and the leaf blade were taken from plants grown in baskets 30 d after sowing. Pulvinus and spikelet samples were taken 54 d after sowing and 1 d before flowering, respectively. Data are shown as mean ± s.d.; $n = 3$ biological repeats. *P* values are based on Student's *t* tests. (**b**–**f**) *In situ* hybridization of *DRO1* mRNA in the root tip of a Dro1-NIL plant. Scale bars, 200 μm. (**b**–**e**) Transverse sections. (**f**) Longitudinal section showing the locations of sections **b**–**e** (arrowheads). (**b**) Elongation zone. (**c**) Distal elongation zone. (**d**) Root apical meristem. (**e**) Region near root cap. Root region identities were estimated from the results shown in **Supplementary Figure 15e**,**f**. (**g**) Relationship between *DRO1* expression level and the ratio of deep rooting 42 d after sowing. V, vector control; KPg-sc and KPg-mc, transformed plants containing a single copy or multiple copies, respectively, of the *DRO1* genomic fragment from KP (*DRO1-kp*); RNAi, plants containing an RNAi knockdown cassette for *DRO1*.

angles than the roots of IR64, vector control lines and RNAi knockdown lines (**Fig. 2c**,**e** and **Supplementary Figs. 9c** and **10c**). These results suggest that *DRO1-kp* confers greater gravitropic curvature than *DRO1-ir* and that the increased gravitropic response in Dro1-NIL plants relative to IR64 plants was caused by their gain of DRO1 function.

DRO1 expression in Dro1-NIL plants was more than double that in IR64 plants in the root tip and the basal part of the shoot (**Fig. 3a**). This difference could be due to transcriptional or post-transcriptional regulation (for example, nonsense-mediated decay), and further analysis will be needed to determine which mechanism is responsible. *DRO1* was expressed around the root apical meristem in the root tip and crown root primordia in the basal part of the shoot (**Fig. 3b**–**f** and **Supplementary Fig. 11**), suggesting that *DRO1* is expressed mainly around root meristem tissues. Although *DRO1* was also expressed in the spikelet, no marked phenotypic difference was observed between the spikelets of IR64 and Dro1-NIL plants (**Fig. 3a** and **Supplementary Fig. 12**). We examined the relationship between *DRO1* expression and root growth angle using *DRO1* transgenic plants. There was a positive relationship between *DRO1* expression and the ratio of deep rooting (**Fig. 3g**), demonstrating that enhancement of *DRO1* expression can increase the root growth angle, resulting in deeper rooting.

The phytohormone auxin has a key role in root gravitropism[16,17], so we examined the effect of auxin on *DRO1* expression. *DRO1* expression was substantially decreased within 30 min of treatment with auxin (2,4-dichlorophenoxyacetic acid or 2,4-D), whereas the expression of *OsIAA20* (an auxin-response marker gene[18]) was increased (**Fig. 4a**). We also investigated the effect of the protein synthesis inhibitor cycloheximide on auxin-dependent repression of *DRO1*. *DRO1* expression was inhibited by auxin, even in the presence of cycloheximide, suggesting that *de novo* protein synthesis is not required for *DRO1* repression by auxin (**Supplementary Fig. 13**). Auxin-response elements (AuxREs), which contain the motif TGTCTC, are found in the promoters of some early-auxin-response genes, and auxin-response factors (ARFs) bind AuxREs to regulate the transcription of these genes[19]. Within AuxREs, the TGTC motif makes the greatest contribution to ARF binding strength[20]. We identified one TGTCTC motif (RE1) and two TGTC motifs (RE2 and RE3) in the *DRO1* promoter region (**Fig. 1b**). To examine whether ARF proteins would interact with AuxREs in this region, we performed electrophoretic mobility shift assays. The recombinant rice ARF protein OsARF1/OsARF23 (homologous to *Arabidopsis thaliana* ARF1, a repressor of AuxRE-mediated gene expression)[21,22] bound to the RE1-containing fragment 1 (**Figs. 1b** and **4b**). This finding suggests that *DRO1* is an early-auxin-response gene that might be directly regulated by ARFs in the auxin signaling pathway. We also performed transcriptome analysis of auxin-related genes in the root tips of IR64 and Dro1-NIL plants before and after treatment with auxin. Expression patterns in response to auxin were similar in the two lines for most of the auxin-related genes tested (**Supplementary Fig. 14**), suggesting that *DRO1* is not involved in the regulation of these genes.

In the horizontal roots of IR64 and Dro1-NIL plants (rotated 90° from the original vertical axis), accumulation of *DRO1* transcripts in the outer cells of the distal elongation zone (DEZ) was less on the lower side than on the upper side of the roots, whereas there was no differential expression near the root cap (**Fig. 4c**,**d** and **Supplementary Fig. 15**). Conversely, expression of *OsIAA20* in the DEZ of both lines was less on the upper side than on the lower side of the roots (**Supplementary Fig. 16**), suggesting that *DRO1-ir* does not interfere with auxin transport. In Dro1-NIL plants, the lower side of horizontal roots showed a 5.4% shorter outer perimeter than the upper side 1.5 h after horizontal rotation (**Fig. 4e**,**f**). This difference

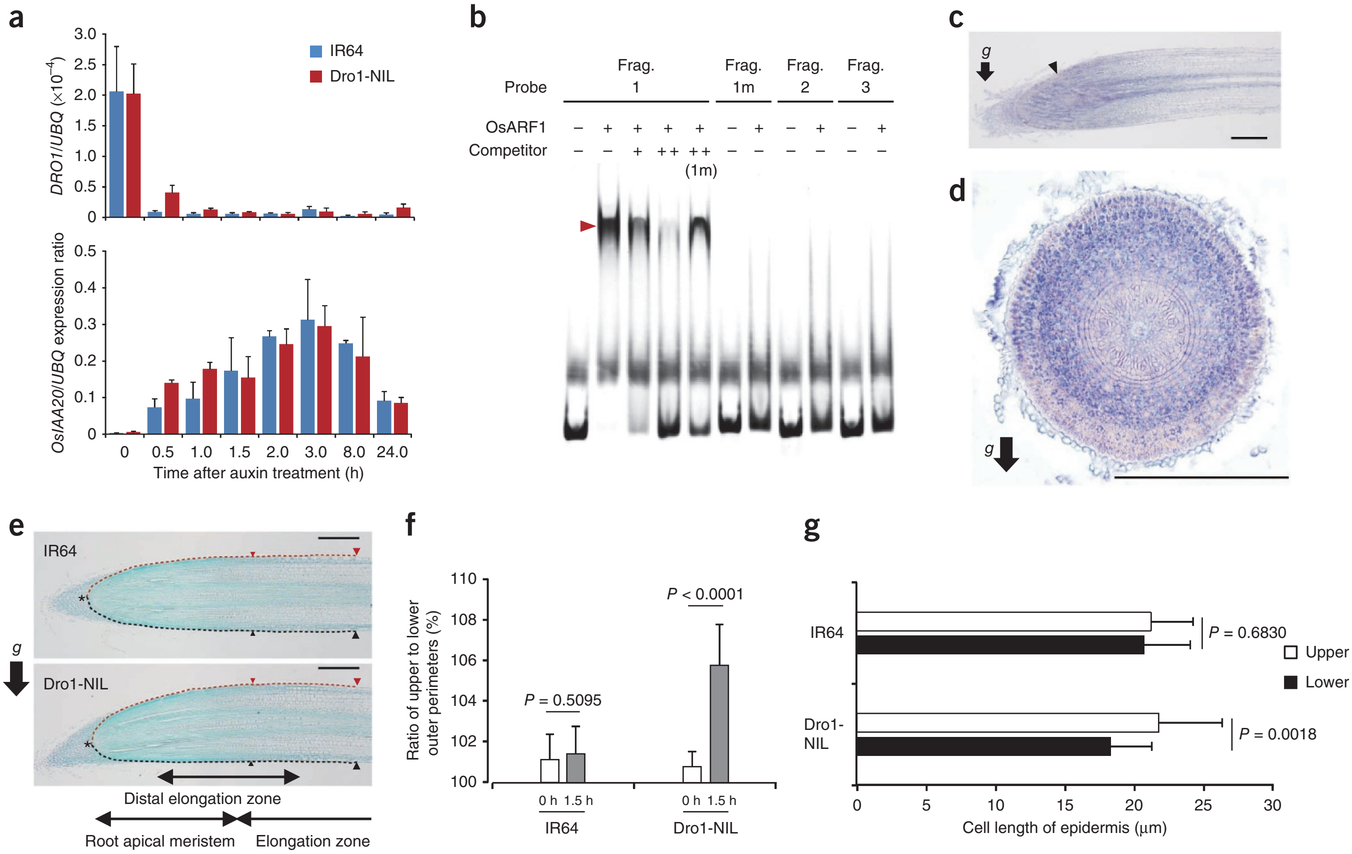


Figure 4 Molecular characterization of *DRO1*. (**a**) Expression of *DRO1* and *OsIAA20* in response to auxin. Seedling root tips of IR64 and Dro1-NIL plants were treated with 10 µM 2,4-D. Data are shown as mean ± s.d.; $n = 3$ biological repeats (15 seedlings per repeat). (**b**) Binding of recombinant OsARF1 protein to AuxRE sequences in the *DRO1* promoter region. The arrowhead indicates the shifted band. The locations of fragments 1, 1m, 2 and 3 are shown in **Figure 1b**. Fragment 1m contained a single-nucleotide substitution (G to A) relative to fragment 1. Competitor was unlabeled fragment 1 (or fragment 1m, where designated). (**c**,**d**) *In situ* hybridization of *DRO1* mRNA in the root tip of Dro1-NIL plants 1.5 h after horizontal rotation. Seedlings were grown on agarose for 2 d. The arrow indicates the direction of gravitational force after rotation. Scale bars, 200 µm. (**c**) Longitudinal section; the arrowhead indicates the position of the transverse section in **d** (distal elongation zone). (**d**) Transverse section. (**e**) Root bending of representative plants (24 plants in each line) 1.5 h after horizontal rotation. Red and black dashed lines indicate outer perimeters from quiescent centers (asterisks) to elongation zone (large arrowheads) on the upper and lower side of roots, respectively. Scale bars, 200 µm. (**f**) Ratio of upper to lower outer perimeters before (0 h) and after (1.5 h) horizontal rotation. Data are shown as mean ± s.d.; $n = 24$ plants for all samples, except for IR64 at 0 h, for which $n = 22$. (**g**) Comparison of the cell length of the epidermis in the elongation zone on the upper and lower sides of the root 1.5 h after horizontal rotation. The regions measured are the 500-µm segments between the arrowheads on each root surface in **e** (800–1,300 µm from the quiescent center). Data are shown as mean ± s.d.; $n = 23$ plants. P values are based on Student's t tests.

may be due to a 15.9% shorter epidermal cell length on the lower side than on the upper side (**Fig. 4g**), although cell dimensions measured in transverse sections were similar (**Supplementary Fig. 17**). Alternatively, the smaller ratio of upper and lower perimeters for IR64 compared to Dro1-NIL roots may be caused by the much smaller difference in epidermal cell length between the upper and lower sides in IR64 plants, resulting in smaller gravitropic curvature in the root maturation zone (**Fig. 2c**,**d**). Upon gravistimulation, auxin flow is redirected to the lower side of the root tip by auxin transporters, causing differential auxin distribution, asymmetric growth and eventual downward root bending[16,17,23]. In *Arabidopsis thaliana* roots, after gravistimulation, the initial curvature originates in the DEZ[24]. Our results suggest that auxin localization on the lower side of the root tip after horizontal rotation represses *DRO1* expression in the DEZ, resulting in decreased cell elongation relative to the upper side. This process may contribute to asymmetric growth, leading to root gravitropic curvature, although further analysis will be needed to clarify how *DRO1* controls cell elongation.

The rice mutations *CROWN ROOTLESS1* (*CRL1*)/*ADVENTITIOUS ROOTLESS1* and *CRL5*, which are directly regulated by ARFs, impair crown root initiation and gravitropism[25–27]. In addition, some rice mutants with mutations in orthologs of the *Arabidopsis* polar auxin transport genes (for example, *GNOM1* and *PINOID1*), which control gravitropism, show abnormal phenotypes for crown root initiation and gravitropism[28–30]. Unlike such mutants, plants with the *DRO1-ir* did not show abnormal root formation other than reduced gravitropism, and *DRO1* has no similarity to the above rice and *Arabidopsis* genes. Therefore, the product of *DRO1* may function in a mechanistically different way from the products of these genes.

To investigate the effect of *DRO1* on drought avoidance, we compared the grain yields of IR64 and Dro1-NIL plants under upland field conditions with no drought, moderate drought or severe drought (**Fig. 5a**). Under severe drought, physiological damage such as leaf wilting and delayed flowering was more prominent in IR64 than in Dro1-NIL plants (**Fig. 5b** and **Supplementary Fig. 18**). Moderate drought significantly reduced the grain weight per plant in IR64

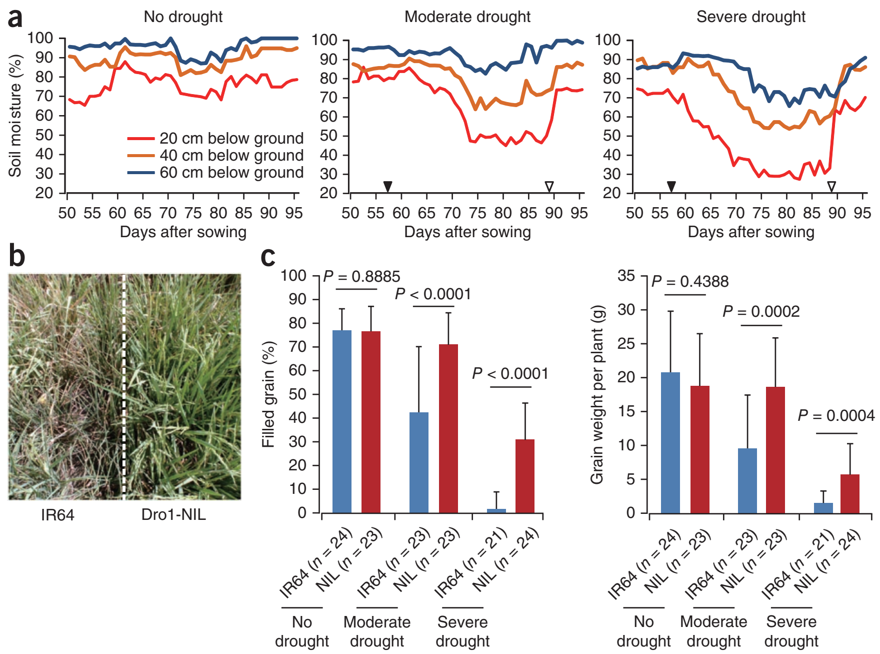


Figure 5 Effect of *DRO1* on the response to drought-induced stress. (**a**) Time course of the moisture level at different soil depths in treatments with no, moderate and severe drought-induced stress under a rainout shelter at CIAT. A soil moisture probe was used to record observations around noon each day in 18 access tubes installed across the experimental area. Black and white arrowheads indicate the time points at which drought conditions were initiated and terminated, respectively. (**b**) Top view of responses of IR64 and Dro1-NIL plants to 27 d of severe drought-induced stress. (**c**) Percentage of filled grain and grain weight for IR64 and Dro1-NIL plants grown under conditions of no, moderate and severe drought-induced stress. Data are shown as mean ± s.d. *P* values are based on ANOVA. NIL, Dro1-NIL.

plants (to 42.3% of that in unstressed plants), whereas Dro1-NIL plants had almost the same grain weight per plant (**Fig. 5c** and **Supplementary Fig. 18**). Under severe drought, the percentage of filled grain in IR64 plants was nearly zero, whereas that in Dro1-NIL plants was >30% (**Fig. 5c**). In the no-drought plots, both lines had similar yields. In a second drought experiment, we measured the photosynthetic capability, root distribution and yield potential of both lines under more severe drought conditions than in the severe-drought plots of the first experiment (**Supplementary Fig. 19**). Dro1-NIL plants had lower leaf temperature (an index of drought resistance[31]) and higher photosynthetic capability than IR64 plants (**Supplementary Figs. 20a**,**b** and **21a**–**c**). The drought-induced flowering delay was shorter in Dro1-NIL plants than in IR64 plants (**Supplementary Figs. 20c**,**d** and **21d**). Consequently, the final grain weight per plant was significantly higher in Dro1-NIL plants than in IR64 plants (**Supplementary Fig. 21e**–**i**). Soil monolith sampling showed that Dro1-NIL plants had more root mass in deeper soil than IR64 plants (**Supplementary Fig. 21j**,**k**). To clarify the relationship between the root's ability to access soil water and drought avoidance, we evaluated the grain yields of IR64 and Dro1-NIL plants in a third drought experiment, under a drought condition in which capillary flow of water from lower to upper soil layers was blocked by a layer of gravel (**Supplementary Fig. 22**). Some roots of Dro1-NIL plants grew through the gravel into the lower soil layer, but those of IR64 plants did not (**Supplementary Fig. 23a**). Dro1-NIL plants thus had higher grain yield than IR64 plants (**Supplementary Fig. 23b**,**c**). When Dro1-NIL and IR64 seedlings were grown in hydroponic culture under osmotic stress imposed by polyethylene glycol (PEG), Dro1-NIL plants were not visibly more tolerant than IR64 plants, suggesting that the two lines have similar dehydration tolerance (**Supplementary Fig. 24**). These results confirm that, under drought conditions, deeper rooting by *DRO1* facilitates better photosynthesis and grain filling, resulting in higher yield.

Sequence analysis of the *DRO1* transcribed regions in 64 rice cultivars identified 6 haplotypes (**Supplementary Fig. 25**). The 1-bp deletion in exon 4 of the IR64 allele (**Fig. 1b**) was found in several IR64 progeny lines showing shallow rooting (**Supplementary Fig. 26**) but not in the ancestors of IR64 (**Supplementary Fig. 26**) or in wild rice (**Supplementary Table 1**). These results suggest that the 1-bp deletion occurred during the modern breeding of IR64 and did not originate from ancestral landraces. The wide variation in deep rooting (**Supplementary Fig. 25**) suggests that deep-rooting QTLs other than *DRO1* may exist in rice and that this variation might be derived from sequence variation in the *DRO1* promoter or other *cis*-acting regulatory regions.

The *DRO1-kp* allele enables rice to produce more grains under drought-induced stress. Under non-drought conditions, these plants show no yield penalty because *DRO1* alters only root growth angle and does not decrease either shoot or root biomass. This characteristic of *DRO1* is of benefit for farmers, who seek the highest possible grain yields under both drought and non-drought conditions. Other economically important monocots, such as maize, contain *DRO1* homologs that may be useful for enhancing drought avoidance in other crops. Our results open the way for new breeding strategies using genes influencing root system architecture to develop crop cultivars with high adaptability to drought.

URLs. RAP-DB, http://rapdb.dna.affrc.go.jp/; US National Institutes of Health ImageJ, http://rsbweb.nih.gov/ij/.

METHODS

Methods and any associated references are available in the online version of the paper.

Accession codes. The DNA Data Bank of Japan (DDBJ) accessions for *DRO1* from IR64 and KP are AB689742 and AB689741, respectively. The microarray data have been deposited in the Gene Expression Omnibus (GEO) under accession GSE46255.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

ACKNOWLEDGMENTS

We thank K. Shimamoto (NAra Institute of Science and Technology, NAIST) and E.E. Hood (Arkansas State University) for kindly providing the pANDA vector and *Agrobacterium* (EHA101), respectively. Full-length cDNAs for use as controls in the quantitative RT-PCR experiments were provided by the Rice Genome Resource Center of NIAS. We thank N. Hattori, T. Tanaka, M. Yamanouchi, K. Matsubara, S. Mochizuki, N. Yokotani, T. Mizubayashi, T. Ando, N. Sentoku and A. Tagiri for technical support and advice on molecular analysis; T. Nishikawa, Y. Itai,

M. Takimoto, E. Nakao, H. Oode, R. Matsushita and the staff of the technical support section of NIAS for technical assistance during field evaluation; M. Kondo, A. Oyanagi, T. Izawa, E. Yamamoto, T. Hoshino, J. Tohme, C.P. Martinez, N.T. Perea and A. Fernando for valuable discussion; and K. Shinozaki for critical reading of the manuscript. This work was supported by a grant from the Ministry of Agriculture, Forestry and Fisheries of Japan (QTL-4003 and RBS-2009).

AUTHOR CONTRIBUTIONS

Y.U. designed the experiments, cloned the *DRO1* gene and wrote the manuscript. Y.U., K.S., Y.K., N.H., H.I., H.T. and R.M. performed molecular analysis of *DRO1*. Y.U., K. Ono and N.K. generated transgenic plants. J.W. and T.M. performed BAC analysis. Y.U., S.O., J.R. and T.T. performed field experiments. K.S., M.I., Y.I., Y.N., K. Okuno and M.Y. provided advice on the experiments.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Plant materials. Lowland rice IR64 (IRGC66970) and upland rice KP (IRG23364) were used as shallow- and deep-rooting cultivars, respectively. Both lines were provided by the International Rice Research Institute. To characterize *DRO1*, we developed a near-isogenic line homozygous for the KP allele of *DRO1* (Dro1-NIL; its generation was BC_4F_4) by repeated backcrossing with IR64 and marker-assisted selection to eliminate non-target DNA regions.

Observation of root distribution by the trench method. We investigated the root systems of IR64, KP and Dro1-NIL plants by means of a trench method[32]. Plants were grown in an upland field under well-watered conditions at NIAS (36°1′ N, 140°6′ E) in Tsukuba, Japan, in the summer of 2009. Growth conditions were managed according to the method described previously[10]. At 120 d after sowing, we dug up the soil near hills of these plants and observed their maximum root depth.

Quantification of root distribution by the soil monolith sampling method. To quantify the root distribution of rice plants, we took a soil monolith (30 cm × 30 cm × 5 cm) from the root area of 120-d-old plants using a metal monolith sampler (Proud) according to the improved monolith method[33] (**Supplementary Fig. 2a–c**). Samples were taken from the same field in which we used the trench method for initial observations. Each soil monolith was divided into 12 blocks, as illustrated in **Supplementary Figure 2c**,**d**. The roots in each block were washed, oven dried at 80 °C for 3 d and weighed to obtain root dry weight.

Ratio of deep rooting. For the quantification of deep rooting in rice plants 35 to 42 d after sowing, the ratio of deep rooting was evaluated using open stainless-steel mesh baskets of soil (top diameter of 7.5 cm and depth of 5.0 cm; Proud) to hold each plant, as described previously[10]. We defined the ratio of deep rooting as the number of roots that penetrated the lower part of the mesh (≥50° from the horizontal, centered on the stem of the rice plant) divided by the total number of roots that penetrated the whole mesh. A larger value for the ratio of deep rooting means that a greater proportion of the roots grew downward.

High-resolution mapping of *DRO1*. We mapped *DRO1* between simple sequence repeat (SSR) markers RM24393 and RM7424 in a previous study[10]. We crossed IR64 and KP plants and then performed repeated backcrossing with IR64 plants as the recurrent parent to transfer the region containing *DRO1* into the IR64 line. For the high-resolution mapping of *DRO1*, we used 4,560 BC_3F_2 plants obtained by selfing BC_3F_1 plants heterozygous for the *DRO1* region. We selected BC_3F_2 plants in which recombination had occurred within the region containing *DRO1*. These plants were self-pollinated, and the progeny were used for positional cloning of *DRO1*.

Transformation rescue test. BAC libraries of the IR64 and KP lines were constructed as described previously[34]. One BAC each of IR64 (IR64-8H21) and KP (KinPat-23H08) was selected from the corresponding BAC library. We subcloned an approximately 8.7-kb genomic fragment of KP containing the *DRO1* region from KinPat-23H08 and cloned it into the pPZP2H-lac binary vector[35], resulting in pPZP∷*DRO1-kp*. The plasmid was introduced into *Agrobacterium tumefaciens* strain EHA101. pPZP∷*DRO1-kp* was then transformed into the IR64 line by *Agrobacterium*-mediated transformation of rice callus, using a callus transformation protocol modified from a protocol for immature embryos described previously[36]. Control plants were generated by introducing the empty vector (pPZP2H-lac).

Subcellular localization analysis. cDNA fragments *DRO1-kp* and *DRO1-ir* without the stop codon were generated by PCR from *DRO1-kp* cDNA from root tips using specific primers (**Supplementary Table 2**). The *CaMV35S*∷*DRO1-kp*∷*YFP*, *CaMV35S*∷*DRO1-ir*∷*YFP* and control (*CaMV35S*∷*YFP*) constructs were then generated as described previously[37]. Rice protoplasts were prepared and transfected as described previously[38]. After transfection, rice protoplasts were incubated in the dark at 26 °C. After 16 h, samples were observed under a confocal laser scanning microscope (TCS-SP5, Leica). For protein blot analysis of DRO1 protein derived from rice protoplasts, construction of the *PUbi*∷*myc*∷*DRO1-kp*∷*YFP* and *PUbi*∷*myc*∷*DRO1-ir*∷*YFP* constructs and immunoblotting were performed as described previously[39]. Myc-tagged proteins were detected with My3 antibody to Myc (M192-3S, MBL). A 1:1,000 dilution of the antibody to Myc was used for protein blot analysis. Bands were visualized with the Chemi-Lumi One Ultra detection system (Nakarai).

Evaluation of root gravitropic curvature. We measured the root gravitropic curvature of IR64 and Dro1-NIL plants after rotation of the roots from the normal vertical axis to the horizontal axis. Seedlings were grown for 2 d on 0.4% agarose (Sigma) in the dark at 30 °C and were then rotated. Seedlings were kept in the dark after rotation, and curvature was measured once each hour for 12 h. We also examined the root gravitropic curvature of IR64, Dro1-NIL and *DRO1-kp* transgenic plants after gravistimulation at different angles. Half of the seedlings from each line were rotated 60° from the original vertical axis for 4 h, and the other half were rotated 90° from the original vertical axis for 4 h. Data from each rotation angle were averaged for each line. We used two different angles of rotation because the original growth angles of the seminal roots in IR64 and Dro1-NIL plants were different (**Fig. 2a**,**b**), and we wanted to examine the relationship between the inclination of the root immediately after rotation and the amount of root curvature 4 h after rotation. Root curvature was measured using US National Institutes of Health (NIH) ImageJ software (see URLs).

Knockdown of *DRO1* expression by RNAi. To produce a construct for the knockdown of *DRO1* (pANDA-*DRO1*), we amplified a 346-bp fragment from exon 3 by PCR using *DRO1*-specific primers (**Supplementary Table 2**), subcloned it into pENTR/D-TOPO (Invitrogen) and transferred it into the RNAi vector pANDA[40]. This plasmid (pANDA-*DRO1*) was then introduced into *Agrobacterium* strain EHA101. pANDA-*DRO1* was introduced into Dro1-NIL by *Agrobacterium*-mediated transformation of rice callus[36].

RNA isolation and expression analyses. Total RNA was isolated from various tissues of IR64 and Dro1-NIL plants using an RNeasy Plant Mini kit (Qiagen). First-strand cDNA was synthesized using SuperScript II reverse transcriptase (Invitrogen). Quantitative RT-PCR using TaqMan probes was performed as described previously[41] using *DRO1*-specific primers and probes (**Supplementary Table 3**). Quantitative RT-PCR using THUNDERBIRD SYBR qPCR Mix (Toyobo) was performed using *OsIAA20*-specific primers[18] (**Supplementary Table 3**). PCR conditions entailed 30 s at 95 °C followed by 40 cycles of 15 s at 95 °C and 1 min at 60 °C. The *OsIAA20* gene was chosen as an auxin-response marker that is expressed in the root tip[42]. All assays were repeated at least three times. To examine the effect of exogenous auxin treatment on *DRO1* expression in root tips (2–3 mm), we submerged the roots of 5-d-old seedlings in a solution of 10 μM 2,4-D (Sigma). For the inhibition of protein synthesis, we soaked the roots of 5-d-old seedlings in water containing 10 μM cycloheximide (Wako) for 24 h and then soaked them in water containing 10 μM cycloheximide with either 10 μM 2,4-D in DMSO or an equivalent amount of DMSO alone (control) for 3 h. To investigate *DRO1* expression in root tips after rotation, we grew seedlings on agarose for 2 d and then rotated them 90° from the original vertical axis for 0.5, 1.0 and 1.5 h. After rotation, tissue was obtained separately from the upper and lower sides of horizontal roots, excluding the stele (**Supplementary Fig. 16a**).

***In situ* hybridization analyses.** We collected root tips (3–4 mm in length) from 30-d-old plants grown in pots and basal parts of stems (3–4 mm in length) from 4-d-old seedlings cultured on 0.3% agar (Wako) medium. To investigate *DRO1* expression in root tips after rotation, we collected root tips (3–4 mm in length) from IR64 and Dro1-NIL plants grown on 0.4% agarose (Sigma) for 2 d and then rotated 90° from the original vertical axis for 1.0 and 1.5 h. Tissue fixation, hybridization and immunological detection of the hybridized probes were performed as described previously[43], with minor modifications. Probes were amplified using specific primers (**Supplementary Table 2**) and cloned into pBluescript II SK(+) (Agilent Technologies).

Electrophoretic mobility shift assays. Four 70-bp *DRO1* promoter fragments, which were synthesized by Klenow fragment (TaKaRa) using oligonucleotide DNA as a template and fragment-specific primers (**Supplementary Table 2**), were cloned into pCRII (Invitrogen). Cy3-labeled probes were amplified using primers designed for pCRII vector sequences (**Supplementary Table 2**). Recombinant OsARF1 protein synthesis and DNA binding reactions were performed as described previously[25,27], with some modifications. Competition experiments were performed using increasing molar amounts (5× and 25×) of unlabeled fragments 1 and 1m. After electrophoresis, Cy3-labeled DNA fragments in the gel were detected with a Typhoon 9400 imager (Amersham Biosciences).

Microarray analysis. To perform microarray analysis, we used RNA isolated from IR64 and Dro1-NIL seedlings 0 and 3 h after treatment with 2,4-D. These treatments and isolation of RNA are described in the Online Methods under 'RNA isolation and expression analyses'. We performed transcriptome analysis with the Rice 4 × 44K Microarray RAP-DB (Agilent Technologies), which contains 35,760 independent probes corresponding to 27,201 loci published in RAP-DB[44,45]. Fluorescent probe labeling and hybridization were performed according to the manufacturer's instructions. Slides were scanned on an Agilent G2505C DNA microarray scanner. Scanned images were analyzed with Feature Extraction Software v. 10.5.1.1 (Agilent Technologies), using default parameters, to obtain background-subtracted and spatially detrended processed Cy3 and Cy5 signal intensities. In this study, 23,535 probes (corresponding to 17,822 loci) gave raw signal intensity above 100 in at least 1 of the 2 samples (IR64 and Dro1-NIL; mean of 3 replicates). We performed *t* tests against zero using the log ratio of signal intensity ($\log_2$ Cy5/Cy3) with the GeneSpring GX12.5 program; *P* values were adjusted for multiple testing by the Benjamini-Hochberg method for correcting for the false discovery rate (FDR). To construct the heatmap, we selected *DRO1* and auxin-related genes[28–30,46,47].

Yield performance test under varying drought-induced stress conditions. To investigate the effect of *DRO1* on yield performance in response to drought-induced stress, we grew IR64 and Dro1-NIL plants in a simulated upland condition under a rainout shelter at CIAT (3°30′ N, 76°21′ W) in Palmira, Colombia, in the summer of 2011. We configured the boom irrigation system (Cherry Creek Systems) inside the rainout shelter to impose conditions of severe, moderate or no stress from drought. The experiment was laid out in a randomized block design with three replications. Drought treatments were imposed from panicle initiation (57 d after sowing) to heading (88 d after sowing). The severe-drought plot received no irrigation after panicle initiation. The moderate-drought plot received irrigation equivalent to 2.5 mm/d of water, and the control plot (no drought-induced stress) received irrigation every day to maintain soil moisture at field capacity. Soil moisture was recorded by soil moisture probes (AquaPro Sensors) every day at 3 depths (20, 40 and 60 cm) in each of 18 access tubes that were installed across the experimental plots 10 d before the initiation of drought treatment. Care was taken to avoid any other stresses (biotic or abiotic) during the experiment. After the termination of the stress treatment (89 d after sowing), soil moisture was maintained at field capacity in all plots. At harvest, a total of 24 plants (8 plants per block) were selected randomly from each treatment, excluding plants in the border rows, to measure grain yield and its components, although plants with rot or insect damage or weakness at the early seedling stage were omitted from samples. We defined the filled grain percentage as the number of filled grains per panicle divided by the sum of filled and unfilled grains per panicle.

Evaluation of rooting depth and yield performance under drought-induced stress. We examined the effects of *DRO1* on root distribution, photosynthetic capacity and grain yield under drought-induced stress conditions. IR64 and Dro1-NIL plants were grown in an upland field at NIAS in Tsukuba, Japan, in the summer of 2010. A randomized block design with three replications was used. Before drought treatment, soil water potential was maintained at greater than −0.02 MPa (well-watered conditions). Soil water potential at a depth of 40 cm was monitored with a tensiometer (UIZ-SMT, Uizin). To impose drought-induced stress, we withheld irrigation from the time of panicle initiation (85 d after sowing) to heading (124 d after sowing). After stress treatment, soil moisture was returned to that in well-watered conditions (greater than −0.02 MPa) by resumption of irrigation. During drought-induced stress treatment, leaf temperature and photosynthesis were simultaneously evaluated between 12 pm and 2 pm on clear days, as described previously[48]. Leaf temperature was measured with a portable infrared thermal imager (TH9100, NEC Avio Infrared Technologies) with a temperature resolution of 0.08 °C. Thermal images of IR64 and Dro1-NIL plants were simultaneously recorded. Leaf temperatures were calculated for each plot isolated in the image using commercially available image-processing software (NS9200, NEC Avio Infrared Technologies). Leaf photosynthesis was measured with a portable photosynthesis system (LI-6400, LI-COR). After plants reached physiological maturity, we randomly sampled 72 plants (24 plants per block, excluding plants in the borders) to measure grain yield and its components and 9 plants (3 plants per block) to measure root traits.

Test of drought avoidance under an artificial drought condition created by blocking capillary water flow. To clarify the relationship between the ability of the root to access soil water and drought avoidance, we set up an assay system inside a greenhouse at NIAS in Tsukuba, Japan, in the summer of 2009 (**Supplementary Fig. 22a**,**b**) according to the method described previously[49]. To construct the drought plot, a 5-cm layer of gravel, contained in coarse netting, was placed on the bed soil followed by a 25-cm layer of soil. The purpose of the gravel was to block capillary water flow from the bed soil to the upper soil layer without inhibiting root growth. The no-drought plot consisted of a 30-cm layer of soil on top of the bed soil, with no gravel layer in between. In the drought plot, the upper soil layer became dry enough to represent a drought condition once we stopped flood irrigation (**Supplementary Fig. 22c**). This assay system can evaluate whether rice plants avoid drought-induced stress by absorbing water from the bed soil through deep roots that have penetrated the gravel layer[49]. The soil type and fertilizer application were the same as described previously[10]. Before drought treatment, soil water potential was maintained at greater than −0.01 MPa (well-watered condition). Soil water potentials at depths of 25 cm (upper soil layer) and 40 cm (bed soil) were monitored with a tensiometer. To impose drought-induced stress, we withheld irrigation from the vegetative stage (58 d after sowing) to heading (106 d after sowing). After heading, soil moisture was returned to that in well-watered conditions (greater than −0.01 MPa) by resumption of irrigation. Each line was represented by 36 hills per plot. Three seeds were sown in each hill (with 20 cm between plants within a row and 40 cm between rows), and plants were thinned to one per hill after seedling establishment. Sixteen plants per line were harvested from each plot, excluding the border rows to avoid edge effects, to measure shoot- and yield-related traits.

Test of dehydration tolerance at the seedling stage in hydroponic culture using polyethylene glycol. Through the use of a hydroponic culture system with PEG as an osmoregulator, the dehydration tolerance of plants can be evaluated under uniform osmotic stress, independent of the effects of root mechanisms for drought avoidance[50,51]. We used this system to test the dehydration tolerance of IR64, Dro1-NIL and KP plants at the seedling stage. Rice seeds were sterilized with 70% ethanol, imbibed in 2% Plant Preservative Mixture (Plant Cell Technology) solution in the dark at 15 °C for 1 d and at 30 °C for 2 d, and then transferred onto a nylon net floated in a small plastic pot filled with 240 ml of 0.25× Kimura B solution[9] (pH 5.5) in a growth chamber (16-h light period at 28 °C and 8-h dark period at 24 °C). The nutrient solution was replaced every 2 d. At 8 d after transplanting, osmotic stress was induced by replacing the nutrient solution with 0%, 10%, 20% or 30% PEG 6000 (Wako). After 2 d of PEG treatment, the plants in each pot were visually scored for osmotic stress damage, which included leaf rolling and wilting, according to the method described previously[50]. Dehydration-tolerance scores were indicated on a scale of 1 (tolerant of osmotic stress) to 9 (intolerant), with possible intermediate scores of 3, 5 or 7. All experiments were performed at least twice, with 20 seedlings per replicate.

Sequencing of *DRO1* transcribed regions in rice accessions. The genomic sequences corresponding to the transcribed regions of *DRO1* were amplified by PCR with 12 primer sets (**Supplementary Table 2**). PCR products were sequenced with an automated fluorescent laser sequencer and the BigDye Terminator v.3.1 Cycle Sequencing kit (Life Technologies).

Statistical methods. All statistical analyses were performed in JMP v. 7.0 software (SAS Institute). The traits evaluated in the field or greenhouse are summarized in **Supplementary Table 4**.

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